In [10]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
# Load real dataset
df = sns.load_dataset("iris")
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
# ====================================================
# A. INDEPENDENT t-TEST
# Species vs categorized sepal
# length
# Petal length across species
# ====================================================
setosa = df[df["species"] == "setosa"]["sepal_length"]
versicolor = df[df["species"] == "versicolor"]["sepal_length"]
t_stat, p_value = stats.ttest_ind(setosa, versicolor)
print("\nINDEPENDENT t-TEST")
print("t-statistic:", t_stat)
print("p-value:", p_value)
if p_value < 0.05:
    print("Reject H0: significant difference.")
else:
    print("Fail to reject H0: no significant difference.")
# ====================================================
# B. CHI-SQUARE TEST
# ====================================================
df["sepal_length_category"] = pd.cut(
    df["sepal_length"],
    bins=[0, 5, 6, 10],
    labels=["Short", "Medium", "Long"]
)
table = pd.crosstab(
    df["species"],
    df["sepal_length_category"]
)
print("\nCONTINGENCY TABLE")
print(table)
chi2, p_value, dof, expected = stats.chi2_contingency(table)
print("Chi-square:", chi2)
print("Degrees of freedom:", dof)
print("p-value:", p_value)
if p_value < 0.05:
    print("Reject H0: significant association.")
else:
    print("Fail to reject H0: no significant association.")
# ====================================================
# C. ONE-WAY ANOVA
# ====================================================
setosa_petal = df[df["species"] == "setosa"]["petal_length"]
```

```
versicolor_petal = df[df["species"] == "versicolor"]["petal_length"]
virginica_petal = df[df["species"] == "virginica"]["petal_length"]
f_stat, p_value = stats.f_oneway(
    setosa_petal, versicolor_petal, virginica_petal
)
print("\nONE-WAY ANOVA")
print("F-statistic:", f_stat)
print("p-value:", p_value)
if p_value < 0.05:
    print("Reject H0: at least one mean differs.")
else:
    print("Fail to reject H0: no significant difference.")
# ======================================================
# D. TUKEY HSD
# ======================================================
print("\nTUKEY HSD")
tukey = pairwise_tukeyhsd(
    endog=df["petal_length"],
    groups=df["species"],
    alpha=0.05
)
print(tukey)
# ======================================================
# E. VISUALIZATION
# ======================================================
plt.figure(figsize=(8, 5))
sns.boxplot(x="species", y="petal_length", data=df)
plt.title("Petal Length by Iris Species")
plt.xlabel("Species")
plt.ylabel("Petal Length")
plt.show()
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
(150, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.057333      3.758000     1.199333
std        0.828066     0.435866      1.765298     0.762238
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000

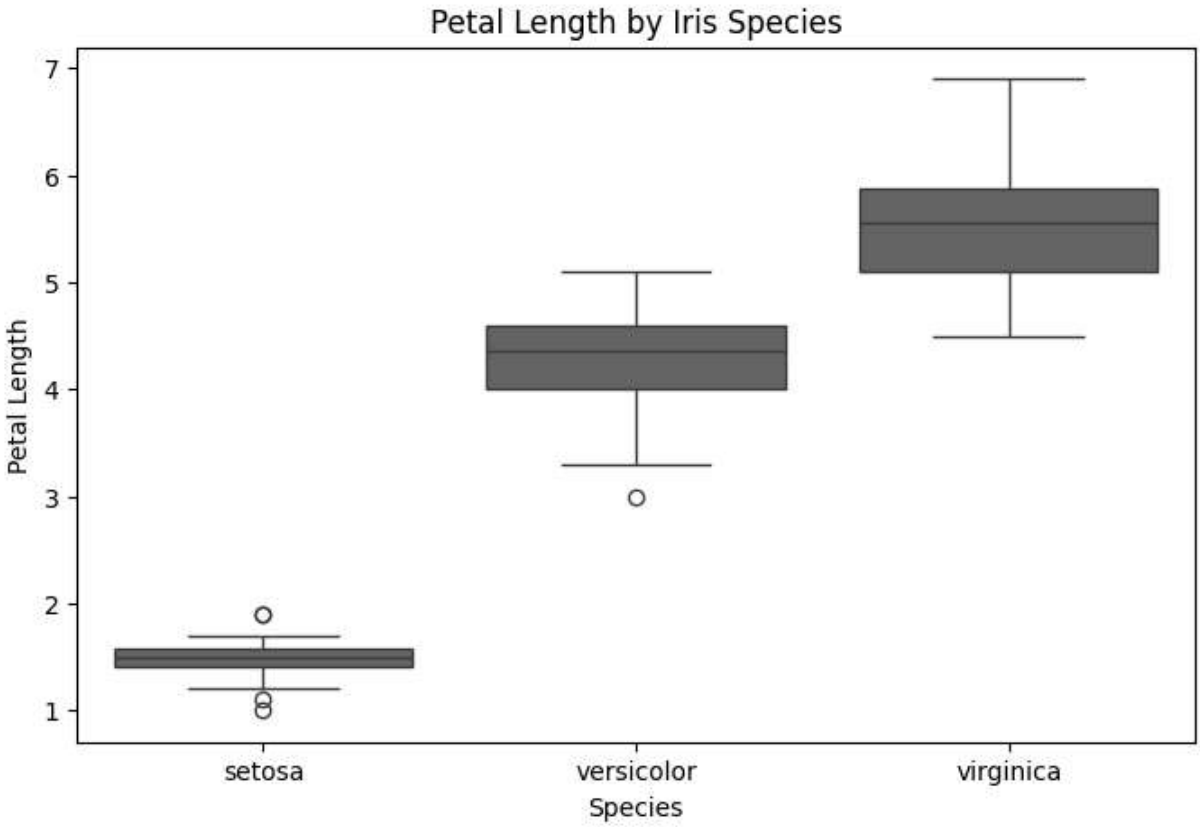
INDEPENDENT t-TEST
t-statistic: -10.52098626754911
p-value: 8.985235037487079e-18
Reject H0: significant difference.

CONTINGENCY TABLE
sepal_length_category  Short  Medium  Long
species
setosa                    28      22     0
versicolor                 3      27    20
virginica                  1       8    41
Chi-square: 93.99228861087146
Degrees of freedom: 4
p-value: 1.8665524919257436e-19
Reject H0: significant association.

ONE-WAY ANOVA
F-statistic: 1180.1611822529785
p-value: 2.8567766109619814e-91
Reject H0: at least one mean differs.

TUKEY HSD
   Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
  group1     group2   meandiff p-adj lower  upper  reject
-----------------------------------------------------
```

```
    setosa versicolor    2.798   0.0 2.5942 3.0018   True
    setosa  virginica     4.09   0.0 3.8862 4.2938   True
versicolor  virginica    1.292   0.0 1.0882 1.4958   True
-----------------------------------------------------------
```



In [8]:
```
!jupyter nbconvert --to html /content/smld_4_new.ipynb
```

```
[NbConvertApp] Converting notebook /content/smld_4_new.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 4 image(s).
[NbConvertApp] Writing 406135 bytes to /content/smld_4_new.html
```